

Yumaa Hidayat

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EDUCATION

Tohoku University, Sendai, Japan

Oct. 2022 – Present

Bachelor of Engineering in Mechanical and Aerospace Engineering

- Specialized in Neurorobotics | Current GPA: 3.46/4.00 | Expected Graduation: September 2026
- Secured the MEXT Scholarship, a four-year fully-funded merit scholarship by the Japanese Government

SMA Negeri 3 Semarang, Semarang, Indonesia

July 2017 – May 2020

Natural Science Course

RESEARCH & PROJECT EXPERIENCE

Undergraduate Thesis

Dec. 2025 – Present

A Simulation Study on the Swimming Performance of a Sea Turtle-like Robot

- Developed a physics simulation model in MuJoCo by overhauling multi-body CAD assemblies in Autodesk Inventor, converting them into STL meshes, and defining rigid-body dynamics via MJCF
- Engineered a bio-inspired gait-phase modulator to alter fin kinematics, outperforming the baseline sinusoidal controller by reducing the cost of transport and increasing swimming velocity at identical frequencies
- Designed and trained a Reinforcement Learning control framework using Stable-Baselines3, enabling controllable robot with precise velocity and yaw-rate tracking for autonomous path-following across circular, sinusoidal, and lemniscate trajectories

Design and Drawing II | SolidWorks Final Project

May 2025 – July 2025

Robotic Arm Assembly

- Translated 2D engineering blueprints into 3D CAD models using SolidWorks, implementing custom modifications on individual components to execute motion-conforming robotic arm assembly
- Conducted kinematic and torque sizing calculations to select standard industrial parts, evaluating gear reduction ratios, interaxial belt lengths, and stepper motors to meet specified operational range constraints

BIOMOD | Biomolecular Design Competition

June 2024 – Nov. 2024

Bidirectional Nano Grabbers

- Collaborated within a 7-member team to design a biostructure using Actin and modified Thymosin $\beta 4$ for self-assembling string-like fibers
- Engineered protein sequences via RFdiffusion and performed 3D structural analysis and visualization utilizing AlphaFold3 and PyMOL
- Validated models against lab experiments, leading the team to secure 4 awards, including 1st Runner Up, Gold Project Award, and 3rd place for Best Website and Video

Computer Seminar 1 | Pygame Final Project

Apr. 2024 – May 2024

Catapult-based Basketball Game

- Developed a custom 2D physics engine and collision detection system using vector mathematics—including dot products and velocity reflection—to simulate ball trajectories and stable impact responses at 60 FPS
- Engineered a modular software architecture and state management system with a dynamic difficulty scaling algorithm based on obstacle movement multipliers
- Conducted a comparative analysis of pixel-to-pixel mask collision versus point-line interaction to eliminate velocity anomalies and optimize runtime stability

ORGANIZATIONAL EXPERIENCE

External Affairs Staff at Perhimpunan Pelajar Indonesia Sendai

Sep. 2024 – Aug. 2025

- Negotiated a collaboration with the Indonesian Embassy (KBRI Tokyo) to establish a mobile passport renewal service in Sendai to eliminate travel barriers for the local diaspora
- Initiated a benchmarking initiatives with peer student organizations, facilitating a platform for knowledge transfer

Event Coordinator at Perhimpunan Pelajar Indonesia Sendai

Sep. 2023 – Aug. 2024

- Coordinated the end-to-end execution of semiannual welcome and graduation parties, managing budgeting, venue logistics, and agenda planning for over 100 attendees per event

SKILLS

Programming Languages: Python, C/C++ (Familiar)

Libraries & Tools: NumPy, Matplotlib, MuJoCo (MJCF), Gymnasium, Stable-Baselines3, TensorBoard, Pygame

Software: SolidWorks, Autodesk Inventor

Spoken Languages: Native Bahasa Indonesia, Fluent English, JLPT N3-level Japanese